Block practice · Logarithms, indices and surds

Work through each question type in order. Each section has a worked-example retry and all further practice. Attempt only the part outlined in red.

What a log means: reading values and switching to index form

Surds: simplifying and rationalising the denominator

Log equations: combine into one log, then remove the log

OLevels.com SCHOOL

Practice workbook · What a log means: reading values and switching to index form

Using the definition of a logarithm to read off a value or to turn one log statement straight into a power.

5 question parts: 1 worked-example retry and 4 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-LOG-01

1. Question 8 · part b.i · 1 mark

2021 JUN · Paper 01

Worked-example retry · Attempt part b.i only.

8 Given that n satisfies the equation

$$\log_a n = \log_a 3 + \log_a(2n - 1)$$

(a) find the value of n.

(3)

Given that $\log_p x = 3$ and $\log_p y - 3\log_p 2 = 4$

(b) (i) express x in terms of p,

(1)

(ii) express xy in terms of p.

(4)

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1. Question 8 · part b.i · 1 mark · Working and answer

Working and answer

OLevels.com SCHOOL

2. Question 3 · part a · 1 mark

2019 JUN · Paper 01 (regional R)

Further practice · Attempt part a only.

3 (a) Write down the value of $\log_3 9$

(1)

(b) Solve the equation $\log_3 9t = \log_9\left(\frac{12}{t}\right)^2 + 2$ where $t > 0$

Give your answer in the form $a\sqrt{b}$ where a and b are prime numbers.

(6)

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2. Question 3 · part a · 1 mark · Working and answer

Working and answer

OLevels.com SCHOOL

3. Question 7 · part a · 1 mark

2022 JAN · Paper 01

Further practice · Attempt part a only.

7 (a) Write down the value of $\log_2 16$

(1)

Given that $4 + 2\log_4 x = \log_2 y$

(b) show that $y = 16x$

(4)

(c) Hence solve the equation $4 + 2\log_4 x = \log_2(4x + 5)$

(3)

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3. Question 7 · part a · 1 mark · Working and answer

Working and answer

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4. Question 10 · part i · 2 marks

2025 JUN · Paper 01

Further practice · Attempt part i only.

10 Solve the equation

(i) $\log_4(6y-5)=3$

(2)

(ii) $\log_4(4-3x)^2-\log_2(x^2-5)-3=0$
Show clear algebraic working.
Give your answer to 3 significant figures.

(9)

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P 7 2 8 6 5 A 0 3 2 3 6

4. Question 10 · part i · 2 marks · Working and answer

Working and answer

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5. Question 9 · part a · 2 marks

2024 JUN · Paper 01

Further practice · Attempt part a only.

9 (a) Find the value of a such that $\log_a 8 = \frac{3}{4}$

(2)

(b) Show that

$$3x\log_2 x - 4\log_{16} 8 + 6x\log_4 8 - \log_2 x = \log_2 (8x)^{3x-1}$$

(4)

(c) Hence solve the equation $3x\log_2 x - 4\log_{16} 8 + 6x\log_4 8 - \log_2 x = 0$

(3)

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5. Question 9 · part a · 2 marks · Working and answer

Working and answer

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Practice workbook · Surds: simplifying and rationalising the denominator

Exact surd arithmetic without a calculator: taking square factors out of roots and clearing surds from denominators.

11 question parts: 1 worked-example retry and 10 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-LOG-04

1. Question 1 · part whole · 4 marks

2023 NOV · Paper 01

Worked-example retry · Attempt part whole only.

1

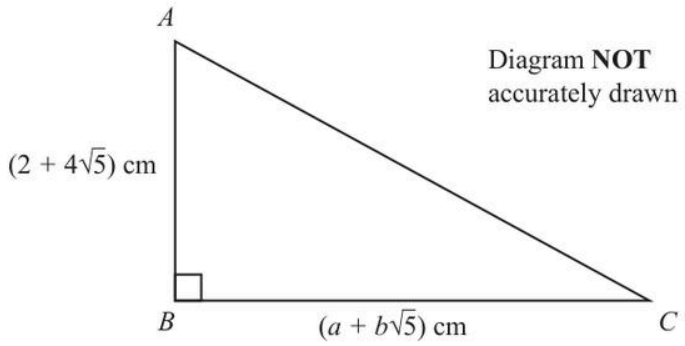


Figure 1

Figure 1 shows the triangle ABC

$\angle ABC = 90°$ $AB = \left(2+4\sqrt{5}\right)$ cm $BC = \left(a+b\sqrt{5}\right)$ cm where a and b are integers.

The area of triangle $ABC = \left(34+11\sqrt{5}\right)$ cm^2

Without using a calculator, find the value of a and the value of b

(4)

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1. Question 1 · part whole · 4 marks

2023 NOV · Paper 01

Worked-example retry · Attempt part whole only.

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Question 1 continued

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(Total for Question 1 is 4 marks)

1. Question 1 · part whole · 4 marks · Working and answer

Working and answer

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2. Question 9 · part c · 2 marks

2022 JAN · Paper 01

Further practice · Attempt part c only.

9 (a) Expand $\frac{1}{\sqrt{1-2x}}$ in ascending powers of x up to and including the term in x^3

Simplify each term as far as possible.

(3)

(b) Show that $\frac{1}{\sqrt{0.96}} = \frac{5\sqrt{6}}{12}$

Show all your working clearly.

(2)

(c) Express $\frac{1}{5\sqrt{6}-12}$ in the form $\frac{a\sqrt{6}}{c} + b$ where a, b and c are integers.

Show all your working clearly.

(2)

Using the expansion you found in part (a) with a suitable value of x

(d) find an estimate, to 5 decimal places, of $\frac{9}{5\sqrt{6}-12}$

(4)

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2. Question 9 · part c · 2 marks · Working and answer

Working and answer

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3. Question 3 · part whole · 5 marks

2025 JUN · Paper 02

Further practice · Attempt part whole only.

3 Given that $\frac{a+b\sqrt{5}}{6-2\sqrt{5}} = \frac{9+4\sqrt{5}}{c}$ where a, b and c are prime numbers,

find the value of a, the value of b and the value of c

(5)

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3. Question 3 · part whole · 5 marks

2025 JUN · Paper 02

Further practice · Attempt part whole only.

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Question 3 continued

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(Total for Question 3 is 5 marks)

3. Question 3 · part whole · 5 marks · Working and answer

Working and answer

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4. Question 1 · part whole · 4 marks

2020 JAN · Paper 01 (regional R)

Further practice · Attempt part whole only.

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1 Given that $\frac{a+\sqrt{3}}{2-\sqrt{3}} = 11 + b\sqrt{3}$ where a and b are integers,

find the value of a and the value of b.

(4)

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(Total for Question 1 is 4 marks)

4. Question 1 · part whole · 4 marks · Working and answer

Working and answer

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5. Question 6 · part a · 2 marks

2022 JUN · Paper 01 (regional R)

Further practice · Attempt part a only.

6 Given that $\dfrac{a+\sqrt{5}}{\sqrt{5}-2} = 11 + 5\sqrt{5}$

(a) without using a calculator, find the value of a
Show your working clearly.

(2)

Triangle PQR is such that

$PR = (x+3)\,\text{cm}$ $QR = x\,\text{cm}$ angle $QPR = 30°$ angle $PQR = 45°$

(b) Show that $x = 3 + 3\sqrt{2}$

(3)

Given that $\sin 105° = \dfrac{\sqrt{6}+\sqrt{2}}{4}$ and that the area of triangle PQR is $A\,\text{cm}^2$

(c) find the exact value of A in the form $\dfrac{9}{8}\left(p\sqrt{6} + q\sqrt{2} + r\sqrt{3} + s\right)$

where p, q, r and s are integers.

(3)

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5. Question 6 · part a · 2 marks · Working and answer

Working and answer

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6. Question 1 · part whole · 3 marks

2022 JUN · Paper 01

Further practice · Attempt part whole only.

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1 Given that $\frac{2\sqrt{3}-4}{3\sqrt{3}+5}$ can be written in the form $a+b\sqrt{3}$ where a and b are integers,

find, without using a calculator, the value of a and the value of b

Show your working clearly.

(3)

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(Total for Question 1 is 3 marks)

6. Question 1 · part whole · 3 marks · Working and answer

Working and answer

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7. Question 1 · part whole · 4 marks

2023 JAN · Paper 01 (regional R)

Further practice · Attempt part whole only.

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1 Given that $\frac{a-\sqrt{48}}{\sqrt{3}+1}$ can be written in the form $b\sqrt{3}-9$ where a and b are integers,

find the value of a and the value of b
Show your working clearly.

(4)

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(Total for Question 1 is 4 marks)

7. Question 1 · part whole · 4 marks · Working and answer

Working and answer

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8. Question 1 · part whole · 3 marks

2025 JUN · Paper 01 (regional R)

Further practice · Attempt part whole only.

1 $\left(8-\sqrt{6}\right)w=50$

Without using a calculator, find the value of w

Give your answer in the form $\frac{a+b\sqrt{6}}{c}$ where a and b are integers and c is prime.

Show your working clearly.

(3)

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(Total for Question 1 is 3 marks)

8. Question 1 · part whole · 3 marks · Working and answer

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9. Question 1 · part whole · 4 marks

2023 JUN · Paper 01 (regional R)

Further practice · Attempt part whole only.

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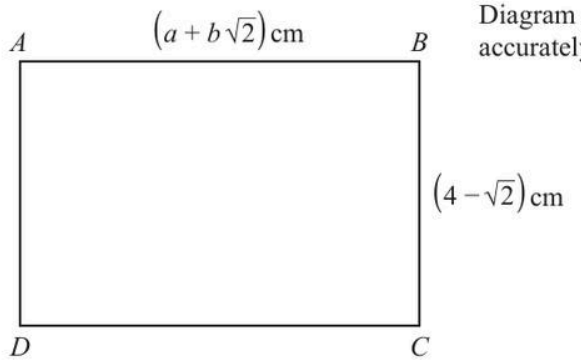


Diagram **NOT** accurately drawn

Figure 1

Figure 1 shows the rectangle $ABCD$.

$AD = BC = (4 - \sqrt{2})$ cm and $AB = DC = (a + b\sqrt{2})$ cm where a and b are integers.

The area of the rectangle is $(10 + \sqrt{2})$ cm^2

Find the value of a and the value of b
Show your working clearly.

(4)

(Total for Question 1 is 4 marks)

9. Question 1 · part whole · 4 marks · Working and answer

Working and answer

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10. Question 9 · part b · 2 marks

2022 JAN · Paper 01

Further practice · Attempt part b only.

9 (a) Expand $\frac{1}{\sqrt{1-2x}}$ in ascending powers of x up to and including the term in x^3

Simplify each term as far as possible.

(3)

(b) Show that $\frac{1}{\sqrt{0.96}} = \frac{5\sqrt{6}}{12}$

Show all your working clearly.

(2)

(c) Express $\frac{1}{5\sqrt{6}-12}$ in the form $\frac{a\sqrt{6}}{c} + b$ where a, b and c are integers.

Show all your working clearly.

(2)

Using the expansion you found in part (a) with a suitable value of x

(d) find an estimate, to 5 decimal places, of $\frac{9}{5\sqrt{6}-12}$

(4)

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10. Question 9 · part b · 2 marks · Working and answer

Working and answer

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11. Question 1 · part whole · 5 marks

2023 JUN · Paper 02

Further practice · Attempt part whole only.

1 Given that $\frac{a+2\sqrt{5}}{3-\sqrt{5}} = \frac{11+b\sqrt{5}}{2}$ where a is an integer and b is prime,

find the value of a and the value of b
Show your working clearly.

(5)

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(Total for Question 1 is 5 marks)

11. Question 1 · part whole · 5 marks · Working and answer

Working and answer

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Practice workbook · Log equations: combine into one log, then remove the log

Log equations in one unknown that are solved by bringing every log to one base (by change of base if needed), combining them with the laws of logs and converting to index form.

16 question parts: 1 worked-example retry and 15 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-LOG-07

1. Question 10 · part whole · 5 marks

2022 JUN · Paper 02

Worked-example retry · Attempt part whole only.

10 Solve the equation

$$\log_4 x + \log_{16} x + \log_2 x = 10.5$$

Show your working clearly.

(5)

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1. Question 10 · part whole · 5 marks

2022 JUN · Paper 02

Worked-example retry · Attempt part whole only.



Question 10 continued

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(Total for Question 10 is 5 marks)

1. Question 10 · part whole · 5 marks · Working and answer

Working and answer

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2. Question 8 · part a · 3 marks

2021 JUN · Paper 01

Further practice · Attempt part a only.

8 Given that n satisfies the equation

$$\log_a n = \log_a 3 + \log_a(2n - 1)$$

(a) find the value of n.

(3)

Given that $\log_p x = 3$ and $\log_p y - 3\log_p 2 = 4$

(b) (i) express x in terms of p,

(1)

(ii) express xy in terms of p.

(4)

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2. Question 8 · part a · 3 marks · Working and answer

Working and answer

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3. Question 8 · part b.ii · 4 marks

2021 JUN · Paper 01

Further practice · Attempt part b.ii only.

8 Given that n satisfies the equation

$$\log_a n = \log_a 3 + \log_a(2n - 1)$$

(a) find the value of n.

(3)

Given that $\log_p x = 3$ and $\log_p y - 3\log_p 2 = 4$

(b) (i) express x in terms of p,

(1)

(ii) express xy in terms of p.

(4)

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3. Question 8 · part b.ii · 4 marks

2021 JUN · Paper 01

Further practice · Attempt part b.ii only.

Question 8 continued

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3. Question 8 · part b.ii · 4 marks

2021 JUN · Paper 01

Further practice · Attempt part b.ii only.

Question 8 continued

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3. Question 8 · part b.ii · 4 marks

2021 JUN · Paper 01

Further practice · Attempt part b.ii only.



3. Question 8 · part b.ii · 4 marks · Working and answer

Working and answer

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4. Question 7 · part b · 3 marks

2022 JUN · Paper 01 (regional R)

Further practice · Attempt part b only.

7 A curve C has equation $y = \log_{10}(x + 2)$

(a) Using the axes below, sketch the graph of C.
Label the coordinates of the points of intersection of C with the coordinate axes.

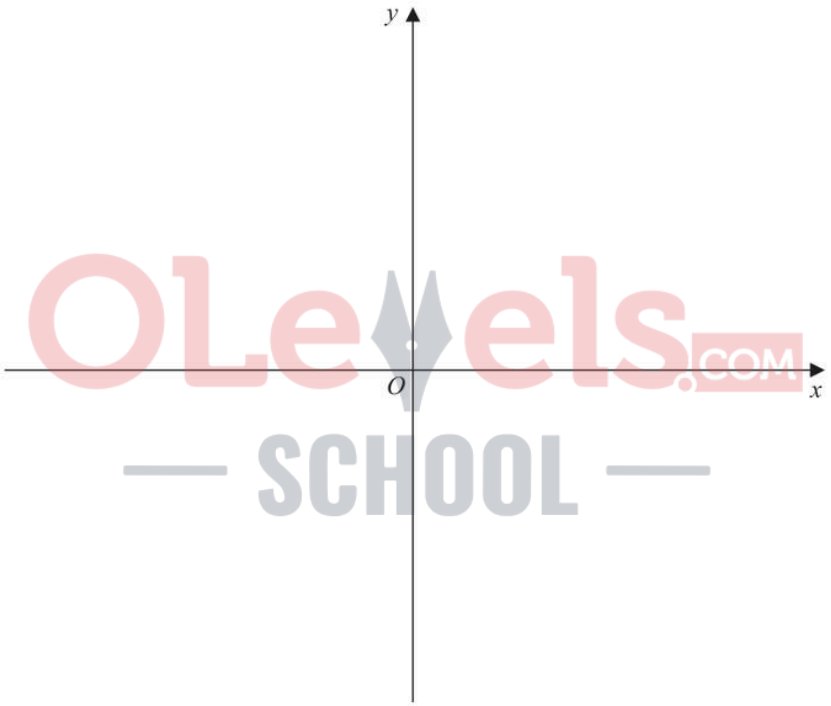
(2)

(b) Solve the equation $2(\log_a 4 + \log_a 16) = 1$

(3)

(c) Solve the equation $5\log_q 16 + 4\log_2 q = 24$

(6)



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4. Question 7 · part b · 3 marks · Working and answer

Working and answer

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5. Question 7 · part c · 3 marks

2022 JAN · Paper 01

Further practice · Attempt part c only.

7 (a) Write down the value of $\log_2 16$

(1)

Given that $4 + 2\log_4 x = \log_2 y$

(b) show that $y = 16x$

(4)

(c) Hence solve the equation $4 + 2\log_4 x = \log_2(4x + 5)$

(3)

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5. Question 7 · part c · 3 marks

2022 JAN · Paper 01

Further practice · Attempt part c only.

Question 7 continued

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(Total for Question 7 is 8 marks)

5. Question 7 · part c · 3 marks · Working and answer

Working and answer

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6. Question 9 · part i · 3 marks

2025 JUN · Paper 02 (regional R)

Further practice · Attempt part i only.

9 (i) Solve the equation $3(\log_a 9 + \log_a 27) = 1$

Give your answer in the form $a = b^c$ where b is prime and c is an integer.

(3)

(ii) Solve the equation $\log_4 p + \log_p 256 = -4$

(6)

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6. Question 9 · part i · 3 marks · Working and answer

Working and answer

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7. Question 3 · part b · 6 marks

2019 JUN · Paper 01 (regional R)

Further practice · Attempt part b only.

3 (a) Write down the value of $\log_3 9$

(1)

(b) Solve the equation $\log_3 9t = \log_9\left(\frac{12}{t}\right)^2 + 2$ where $t > 0$

Give your answer in the form $a\sqrt{b}$ where a and b are prime numbers.

(6)

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7. Question 3 · part b · 6 marks

2019 JUN · Paper 01 (regional R)

Further practice · Attempt part b only.

Question 3 continued

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(Total for Question 3 is 7 marks)

7. Question 3 · part b · 6 marks · Working and answer

Working and answer

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8. Question 7 · part b · 4 marks

2022 JAN · Paper 01

Further practice · Attempt part b only.

7 (a) Write down the value of $\log_2 16$

(1)

Given that $4 + 2\log_4 x = \log_2 y$

(b) show that $y = 16x$

(4)

(c) Hence solve the equation $4 + 2\log_4 x = \log_2(4x + 5)$

(3)

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8. Question 7 · part b · 4 marks · Working and answer

Working and answer

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9. Question 9 · part b · 4 marks

2024 JUN · Paper 01

Further practice · Attempt part b only.

9 (a) Find the value of a such that $\log_a 8 = \frac{3}{4}$

(2)

(b) Show that

$$3x\log_2 x - 4\log_{16} 8 + 6x\log_4 8 - \log_2 x = \log_2 (8x)^{3x-1}$$

(4)

(c) Hence solve the equation $3x\log_2 x - 4\log_{16} 8 + 6x\log_4 8 - \log_2 x = 0$

(3)

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9. Question 9 · part b · 4 marks · Working and answer

Working and answer

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10. Question 9 · part c · 3 marks

2024 JUN · Paper 01

Further practice · Attempt part c only.

9 (a) Find the value of a such that $\log_a 8 = \frac{3}{4}$

(2)

(b) Show that

$$3x\log_2 x - 4\log_{16} 8 + 6x\log_4 8 - \log_2 x = \log_2 (8x)^{3x-1}$$

(4)

(c) Hence solve the equation $3x\log_2 x - 4\log_{16} 8 + 6x\log_4 8 - \log_2 x = 0$

(3)

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10. Question 9 · part c · 3 marks

2024 JUN · Paper 01

Further practice · Attempt part c only.

Question 9 continued

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10. Question 9 · part c · 3 marks

2024 JUN · Paper 01

Further practice · Attempt part c only.

Question 9 continued

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10. Question 9 · part c · 3 marks

2024 JUN · Paper 01

Further practice · Attempt part c only.

Question 9 continued

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(Total for Question 9 is 9 marks)

10. Question 9 · part c · 3 marks · Working and answer

Working and answer

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11. Question 7 · part whole · 5 marks

2020 JAN · Paper 01

Further practice · Attempt part whole only.

7 Solve the equation

$$\log_7(8x^2 - 6x + 3) - \log_{49} x^2 = 3\log_7 2$$

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11. Question 7 · part whole · 5 marks

2020 JAN · Paper 01

Further practice · Attempt part whole only.

Question 7 continued

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(Total for Question 7 is 5 marks)

11. Question 7 · part whole · 5 marks · Working and answer

Working and answer

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12. Question 10 · part ii · 9 marks

2025 JUN · Paper 01

Further practice · Attempt part ii only.

10 Solve the equation

(i) $\log_4(6y-5)=3$

(2)

(ii) $\log_4(4-3x)^2-\log_2(x^2-5)-3=0$
Show clear algebraic working.
Give your answer to 3 significant figures.

(9)

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12. Question 10 · part ii · 9 marks

2025 JUN · Paper 01

Further practice · Attempt part ii only.

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Question 10 continued

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(Total for Question 10 is 11 marks)

12. Question 10 · part ii · 9 marks · Working and answer

Working and answer

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13. Question 5 · part whole · 7 marks

2021 NOV · Paper 01

Further practice · Attempt part whole only.

5 Solve the equation

$$\log_3 \sqrt{x-5} + \log_9 (x+3) - 1 = 0$$

Show clear algebraic working.

(7)

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13. Question 5 · part whole · 7 marks

2021 NOV · Paper 01

Further practice · Attempt part whole only.

Question 5 continued

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(Total for Question 5 is 7 marks)

13. Question 5 · part whole · 7 marks · Working and answer

Working and answer

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14. Question 5 · part a · 2 marks

2020 JAN · Paper 01 (regional R)

Further practice · Attempt part a only.

5 (a) Show that $\log_4 32 = \frac{5}{2}$

(2)

(b) Hence, or otherwise, find the exact solutions of the equation

$$\log_2 x - \log_4 32 + \frac{1}{4}\log_x 16 = 0$$

(7)

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14. Question 5 · part a · 2 marks · Working and answer

Working and answer

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15. Question 6 · part i · 4 marks

2024 NOV · Paper 02

Further practice · Attempt part i only.

6 (i) Solve the equation $5(\log_b 9 + \log_b 3) = 3$

(4)

(ii) Solve the equation $3\log_3 x + 3\log_x 27 = 8\log_4 128$
Give your answers in exact form.

(7)

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P 7 3 5 8 7 A 0 1 4 3 2

15. Question 6 · part i · 4 marks · Working and answer

Working and answer

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16. Question 4 · part whole · 7 marks

2020 NOV · Paper 02

Further practice · Attempt part whole only.

4 (i) Solve the equation $16\log_r 4 = \log_4 r$

(2)

(ii) Solve the equation $\log_5 9 + \log_5 12 + \log_5 15 + \log_5 18 = 1 + \log_5 x + \log_5 x^2$

(5)

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P 6 2 2 8 5 A 0 8 3 2

16. Question 4 · part whole · 7 marks

2020 NOV · Paper 02

Further practice · Attempt part whole only.

Question 4 continued

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(Total for Question 4 is 7 marks)

16. Question 4 · part whole · 7 marks · Working and answer

Working and answer

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